

An Automated Magnetic Micro-Robot Assembly System

Oliver J. Shindell¹ and David J. Cappelleri^{1,2}

Abstract—This paper presents the Automated Magnetic Micro-Robot Assembly System, a novel system that is used for fabricating micro-robots with embedded permanent magnets. A micromanipulator is used to control tweezers, and stepper motors are used to control a system of four stages where different assembly steps are conducted. Vision-based feedback from an overhead microscope camera ensures precise assemblies, employing contour-based image processing algorithms to identify robots, magnets, and other features. Embedded magnets are 250 μ m cubes. A single robot takes approximately 192 seconds to assemble, plus 119 seconds for each additional embedded magnet beyond the first, for more complex designs. These times are 54% and 48% faster than the assembly times for a well-practiced human manually performing the task. A total of 20 μ -magnets can be embedded in a single batch. This work is an advancement in the field of micro-assembly, pushing micro-robotics closer to real applications by establishing a basis for mass production.

Keywords—microassembly, microrobotics, micromanipulation

I. INTRODUCTION

Progress in the field of micro-robotics indicates high potential for minimally invasive medicine and tissue engineering [1][2]. Magnetic micro-robots are favored for untethered, precise, and safe *in vivo* control [3][4]. Additionally, they offer versatility in design and can perform a wide range of tasks [5][6]. Increasing the amount of magnetic material in a micro-robot, and the strength of magnetic material, yields more precise control and a wider range of functionality [7]. An effective way to do this is to embed a strong permanent magnet into the micro-robot body. The size of micro-robots prohibits the use of embedded electronics that broadcast signals, so they must be observed by an external source [8]. An embedded permanent magnet can be distinctly observed by an ultrasound system, making detection more accurate [9].

A popular use of magnetic micro-robots is targeted drug delivery, which is largely due to their small size and high maneuverability [10]. Delivering more drugs in this manner requires that more robots be used once. Magnetic micro-robots are also effective when performing other tasks in large swarms, such as moving larger objects or guiding an organism [11].

Embedding a magnet into a micro-robot requires a delicate and precise pick-and-place operation. This sort of contact-based manipulation is a prominent field of research, with

related work in areas such as assembly [12][13], grasping delicate objects [14], and manipulating magnetic particles [15]. There is also specific research on how to assemble magnetic robots [16]. To assemble large numbers of micro-robots, it is desirable to have a system that works autonomously, with consistency and no human oversight. Presented in this paper is an Automated Magnetic μ -Robot Assembly System (AM μ RAS). It employs electro-mechanical actuators and computer vision to autonomously assemble micro-robots using contact-based manipulation, removing the necessity for human operation and serving as a basis for scaling production.

II. SYSTEM OVERVIEW

The assembly system developed is shown in Fig. 1. A detailed view of the stage subsystem is shown in Fig. 2, with a close-up of the tweezer tips seen in Fig. 2(A). The stepper motor in Fig. 1(A) opens and closes tweezers (Dumont) using a lead-screw system. This motor, along with the four others in the system, is driven by a stepper motor driver circuit. These drivers are controlled by a microcontroller (Arduino). The host computer communicates by serial with the microcontroller, the micromanipulator controller (Sutter MPC-200), and the camera (Basler). The micromanipulator (Sutter MP-285) tracks its own state as steps in XYZ coordinates. The camera is fitted on a 0.75X magnification lens (Edmund Optics), which is fixed

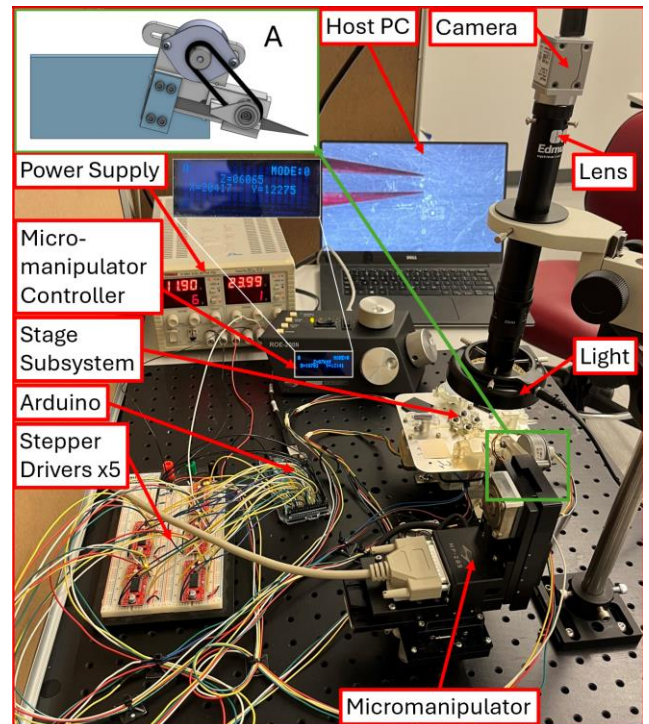


Figure 1. Full labeled view of AM μ RAS. (A). Close-up of tweezer actuation system.

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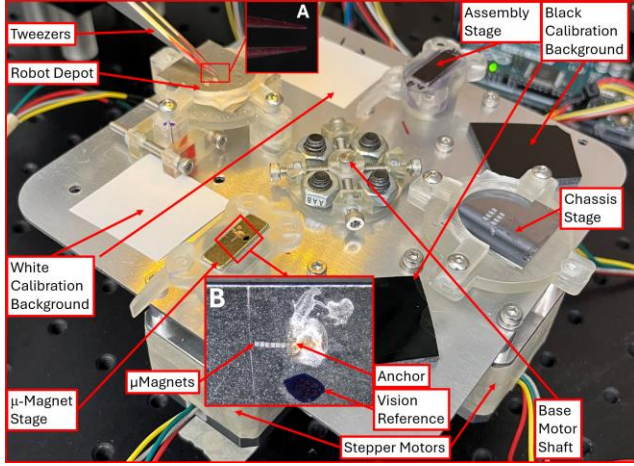


Figure 2. Stage subsystem view. (A). Close-up view of tweezers. (B). Close-up of μ -magnet stage.

at a height such that all micromagnets and microrobot chassis are in the focus plane. The host computer runs a Python script that interacts with all three peripherals to conduct assemblies.

The stage subsystem has eight areas on it, as shown in Fig. 2. The entire system is mounted on a stepper motor, which can position each area under the camera. The black and white calibration backgrounds are used to obtain clear images of the tweezers, which helps relate the state of the micromanipulator to the video feed from the camera. Placing them below the focal length of the lens yields a sharp image of the tweezers. The robot depot is used to store assembled robots. Its surface is a sheet of nickel covered with a small amount of resin, which ensures that robots stick to the depot rather than the tweezers. These areas are all fixed. The chassis stage contains empty chassis. Its surface is a black silicon wafer, which contrasts well with chassis. Both the chassis stage and robot depot have a semi-circular fixture that secures the assembly substrate placed on them. Placing different substrates on these stages can change the height of the surface, so set screws are used to make sure all stages are at the same height.

Both the μ -magnet stage and assembly stage are N52 magnets, poled through their length, that are pressed into SLA printed pieces set-screwed to stepper motors. The μ -magnet stage holds the supply of μ -magnets to be embedded into chassis. Fig. 2(B) shows the features that enable magnets to be isolated from each other; this process is explained in Section 3. The anchor is made with two-photon polymerization [17], with an embedded μ -magnet the same size as those for assembly. It is secured in direct contact with the stage by an adhesive. The surface of the assembly stage is blackened with a marker, to create more contrast with magnets placed on top of it.

The tweezers are fixed at a constant 15° angle with the XY plane. Different tweezers can be removed and replaced. Shown in Fig. 3 are modifications made to tweezers. These modifications are made to improve visibility, or to tweeze chassis that are too fragile, have excessively complex geometry, or are too flexible to be grasped by the tweezer tips. Fig. 3(A) shows a rectangular chassis grasped by tweezer tips. The tips are colored red for high visibility. In Fig. 3(B), pieces made using two photon polymerization are affixed to the tweezer tips. They are tailored to accommodate the chassis to be grasped. Fig. 3(C) shows the grasping of a microrobot that

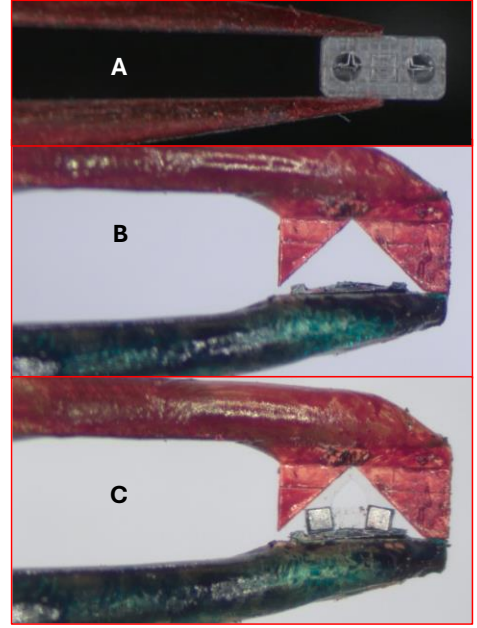


Figure 3. (A) Tweezers holding a tumbling robot chassis. (B) Tweezers with modified tips. (C) Tweezers with modified tips holding an assembled gripper robot.

consists of two sides joined by a flexible spring-like structure. The smooth distribution of pressure mitigates mechanical destruction or deformation of the chassis and ensures that flexible parts are sufficiently constrained. Further information on the micro-robots shown in this figure can be found in [18][19].

III. SYSTEM OPERATION

AM μ RAS requires that materials for assembly be inserted before operation begins. The necessary materials are μ -magnets, chassis, and a nickel substrate for the robot depot. The μ -magnets are loaded onto the μ -magnet stage as needed. This stage begins positioned under the camera, oriented as seen in Fig. 2(A). The assembly stage must be oriented such that the poling direction of the embedded magnet is opposite that of the μ -magnet stage – otherwise μ -magnets will be pulled to its ends. To maximize assembly capacity, chassis should all be printed in the same orientation and centered on the stage.

Manual calibration may be necessary prior to operation. The micromanipulator operates on an XYZ coordinate system of steps, each step being about one micron. The Z state at which tweezers touch each stage must be found, as well as the Z state which gives the sharpest image of the tweezers. Small inaccuracies in these calibrations do not compromise the system. This calibration only needs to be performed if a substrate on the chassis stage or robot depot is replaced with one of a different thickness or if the tweezers are changed.

The micromanipulator must be able to precisely move the tweezers to any location in view of the camera. Thus, a relationship must be established between the micromanipulator's XYZ state and XY coordinates of pixels seen by the camera. Representing the micromanipulator state as (X_M, Y_M, Z_M) and the position of the tweezers in the image as (X_{twz}, Y_{twz}) in pixels, we introduce the constants $X_X, Y_X, X_Y,$

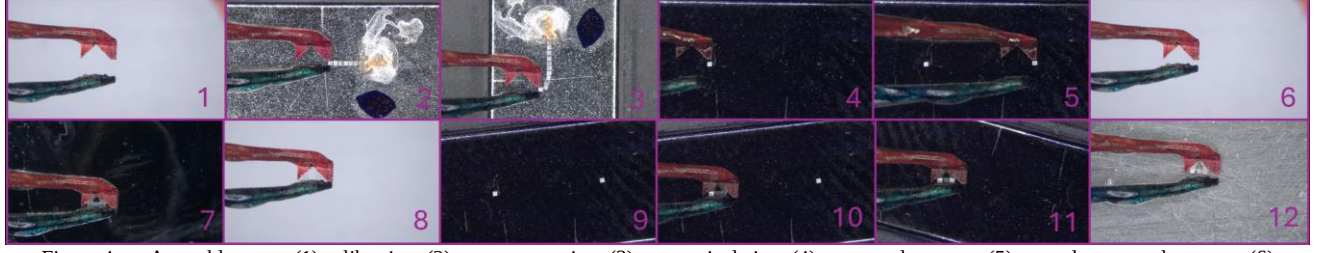


Figure 4. Assembly steps: (1) calibration, (2) magnet grasping, (3) magnet isolation, (4) magnet placement, (5) second magnet placement, (6) calibration, (7) chassis grasping, (8) calibration, (9) magnet alignment, (10) magnet embedding, (11) realignment & second magnet embedding, (12) finished robot release

and Y_Y , which relate change in (X_M, Y_M) to change in (X_{twz}, Y_{twz}) with the following equations:

$$\Delta X_M * X_X + \Delta Y_M * X_Y = \Delta X_{twz} \quad (1)$$

$$\Delta X_M * Y_X + \Delta Y_M * Y_Y = \Delta Y_{twz} \quad (2)$$

The constants that satisfy these two equations are found through a calibration where the movement of the tweezer tip associated with X and Y micromanipulator movement is observed and recorded. This calibration is more accurate than a simple conversion of pixels to SI units of length, and accounts for angular differences between the XY coordinate systems of the camera and micromanipulator. Once performed, it is valid until either the micromanipulator or the camera is moved.

For this calibration to be useful, the X_M and Y_M that move the tweezers to a set location X_{twz}, Y_{twz} must be found as well. Let (X_{cal}, Y_{cal}) be the values of (X_M, Y_M) that position the tweezers in the center of the image, denoted as (X_{cnt}, Y_{cnt}) in pixels. Applying algebra yields the following equations:

$$U_X = X_{des} - X_{cnt} + X_Y * Y_{cal} + X_X * X_{cal} \quad (3)$$

$$U_Y = Y_{des} - Y_{cnt} + Y_Y * Y_{cal} + Y_X * X_{cal} \quad (4)$$

$$X_M = \frac{X_Y * U_Y - Y_Y * U_X}{X_Y * Y_X - X_X * Y_Y} \quad (5)$$

$$Y_M = \frac{X_X * U_Y - Y_X * U_X}{X_X * Y_Y - Y_X * X_Y} \quad (6)$$

X_{des} and Y_{des} are desired coordinates in the image, and U_X and U_Y are variables used to simplify the calculation; they represent no physical quantity. Using these equations, the micromanipulator state that positions the tweezers at a desired location in the image can be found.

The XY position of the tweezer tip is not nearly as consistent as the Z position, and a far greater degree of accuracy is required. Opening and closing the jaw affects the tip position, as does adjusting the grip on a tweezed object. This necessitates re-evaluating X_{cal} and Y_{cal} while the system is operating. This is done by detecting the tweezer tip, a process detailed in Section 4C, and using (5) and (6) to move it to the center of the image. Then, the values of X_{cal} and Y_{cal} are reset to the XY state of the micromanipulator. This process is repeated until the tweezer tip is detected as being in the center of the image. A single iteration suffices for empty tweezers, but if the tweezers are holding an object, the image is noisier, so iteration is necessary for accuracy.

Loaded with the necessary materials and calibrated, the system can begin operation. A series of images showing the operation steps is seen in Fig. 4. In Step 1, the tweezers are calibrated over a calibration background. If the tweezers have

modified tips, a white background is used for all calibrations. The jaw distance is also adjusted to a desired value. In Step 2, a μ -magnet on the μ -magnet stage is grasped, and the μ -magnet is removed from the group in Step 3. This is done by rotating the stage while moving the tweezers such that the μ -magnet group remains in line with and attached to the anchor. Rotating 90° orients the tweezed μ -magnet perpendicular to the others, which allows it to be removed easily. In Step 4, the isolated magnet is placed on the assembly stage. Step 5 is the result of repeating Steps 2-4, placing another μ -magnet on the assembly stage. Once all μ -magnets are placed, Step 6 calibrates the tweezers and again adjusts the jaw opening. In Step 7 a chassis is taken from the chassis stage, after which another tweezer calibration is performed in Step 8. If the tweezers don't have modified tips, this calibration is done by detecting the tweezed chassis rather than the tweezers. After calibration, Step 9 returns to the assembly stage and rotates it to orient a μ -magnet for embedding with the correct magnetic poling direction. In Step 10, the chassis is placed onto the μ -magnet, securing it inside. During this placement, the tweezers descend in a helical pattern, correcting for any small alignment errors. Step 11 repeats this for the second hole, again re-orienting the μ -magnet to ensure it is poled in the proper direction (μ -magnets are poled in opposite directions for the robot shown in the figure). Finally, the completed robot is deposited at the depot in Step 12. If there are more robots to be assembled, the system continues.

Once AM μ RAS finishes, a set of assembled robots is left on the robot depot. The user can remove them individually from the nickel substrate or remove the entire substrate and keep the assembled robots on it. Completed robots should be washed with iso-propyl alcohol to remove excess resin from inside the chassis or the substrate and any coloring that may have stuck on the robots during the assembly. Curing with UV light secures the μ -magnets inside.

Although AM μ RAS is highly accurate, manufacturing defects in the μ -magnets or chassis can cause assembly issues. As such, safety logic is employed to detect and deal with assembly failures when they occur, as shown in Fig. 5. The μ -magnets placed on the assembly stage occasionally move out of the field of view when released from the tweezers, although such a failure is extremely rare, only occurring when Z calibration was incorrect or the assembly stage's embedded magnet was backwards. The μ -magnet embedding may fail if a chassis hole is too small or a μ -magnet is too large. When this happens, the chassis rotates about the Y axis and needs to be released and regripped to fix the orientation. If either a μ -magnet or a chassis fails during embedding twice, AM μ RAS

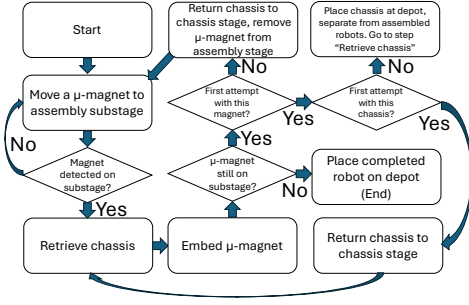


Figure 5. Safety logic flowchart

deems it as defective and discards it. This ensures that AMuRAS works even if a step fails or a part is defective.

IV. COMPUTER VISION & IMAGE PROCESSING

In order to execute the processes described in Section 3, AMuRAS employs a variety of vision algorithms to obtain the positions of magnets, tweezers, and chassis. These algorithms primarily use the CV2 library available in Python. Since the system is highly stable, it is unnecessary to use continuous image feedback. Rather, the host computer will process a single image, and based on that will send instructions to the microcontroller and micromanipulator to complete assembly steps, the only exception being the iterative calibration of the tweezers. Processing a small number of images allows algorithms to run slowly without major drawbacks.

A. Detection of chassis

Two types of chassis detection schemes are used in this system, one for simpler geometry and one for more complex geometry. The simple method begins by taking an image, applying a binary color mask, and finding contours. Contours that fall outside an area threshold are discarded and the remaining contours are approximated as an octagon. An example of this is shown in Fig. 6(A). Two points that approximate one side of the chassis are shown in the same color. When chassis are oriented correctly, the line connecting green points should be horizontal, while that connecting blue points should be vertical. The orientation of a chassis can be calculated based on the angle of these lines.

For more complex geometry, a more complex and robust algorithm is required. The algorithm used by this system is a simplified version of shape matching based on contour approximation. For this algorithm to work, an image of a computer model of the chassis in a desired orientation is used as a reference. The outline of the model is treated as a contour, which is dilated and eroded to simplify the geometry. This simplified contour is then approximated as a polygon, starting with three sides and increasing in number of sides until the approximation error begins to change by only a small amount. After scaling and translation, the points of this polygon serve as reference points for a physical chassis in a desired location and orientation. To an image of one or more physical chassis, a color-based binary threshold is applied and contours are identified and filtered. Then the same steps as applied to the model can be applied to each contour in the image, yielding polygons with the same number of points as the reference polygon.

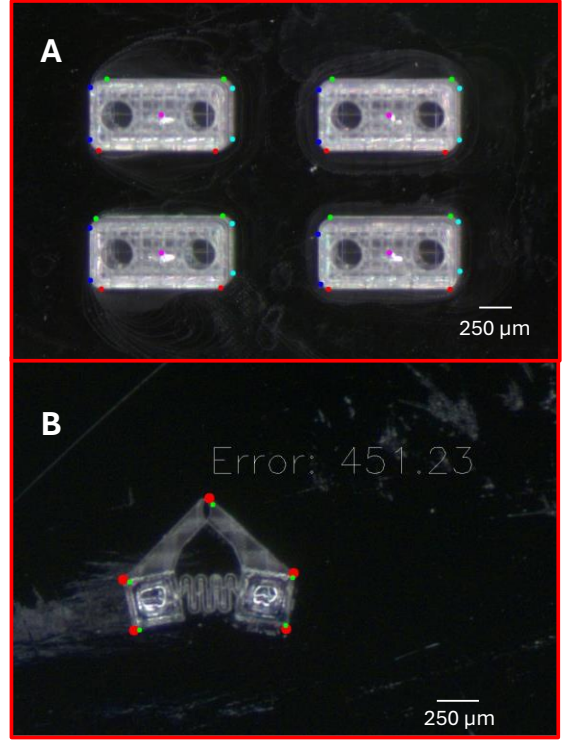


Figure 6. (A) Detection results for tumbling robots. (B) Detection results for gripping robots. Approximation in red, reference in green. Error units are pixels squared.

To obtain information about the position of the real chassis, the reference points must be matched as closely as possible to the approximation of the real chassis. This is done using the gradient descent optimization tool from the SciPy library. An error function is defined as the sum of squared distances between each reference point and its nearest real point. Then, a rotation and translation can be found that minimizes this error function. For a chassis with no radial symmetry there will be as many local minima as there are reference points. To prevent the algorithm from finding a non-global minimum, it iterates as many times as there are reference points after the initial optimization. Each iteration uses the translation found in the first optimization as an initial guess, and a range of initial rotation angles is used, evenly spaced from 0° to 360°. This ensures that at least one repetition will converge towards the global minimum. The resulting optimal rotation and translation represent the rotational and translational deviation from the desired position, with an associated error in units of pixels squared. If optimization fails to converge below a certain error threshold, the detected object can be labeled as not being a chassis. An example of this algorithm is seen in Fig. 6(B), showing the estimated points, reference points, and error.

This detection scheme provides excellent accuracy and robustness, and consistent rejection of foreign bodies and noise. It also requires minimal computing resources and training data, requiring only a computer model and a single image of a real chassis. This makes it a robust detection algorithm for many different varieties of chassis.



Figure 7. Detection results for isolated μ -magnets. Blue crosses denote centroids. The angle shown is that of the leftmost magnet, in degrees.

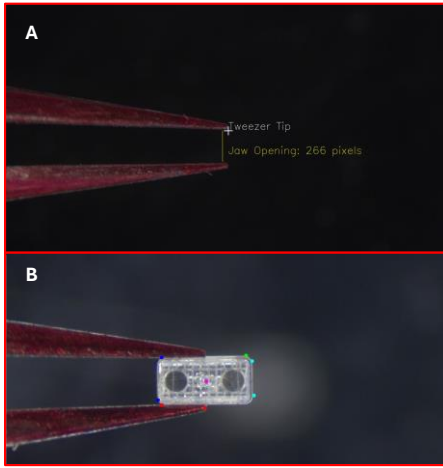


Figure 8. Tweezer detection. (A) Results of tweezer detection for calibration. (B) Estimation of a chassis being grasped.

B. Detection of μ -magnets

AM μ RAS detects μ -magnets in two possible states – grouped with other magnets or isolated. Magnets appear in the grouped state on the μ -magnet stage, and in the isolated state on the assembly stage. Hence, different detection schemes are used for the different stages.

For the assembly stage, the goal is to identify every isolated μ -magnet in view and find their orientations and locations. To do this, a color-based binary threshold is applied to a captured image of the stage with magnets on it. Contours in this image are identified, filtered by area, and dilated and eroded for smoothing. Using CV2 functions, the minimum area rectangle enclosing each contour is found. For a rectangle to be counted as a magnet, it must have sides within a certain size threshold, and it must have a percentage of filled area above a certain threshold. These filters effectively identify μ -magnets' locations and orientations and filter out noise from foreign objects and uneven surfaces. Results are shown in Fig. 7.

On the μ -magnet stage, the goal is to identify the location of the μ -magnet at the end of the group, as the tweezers need only grasp that magnet. The blue mark on the stage serves as an identifier to find the location of the anchor. Since the number of μ -magnets is a known quantity, a mask can be applied to the image to only look at the area where magnets will be. A color-based threshold is applied to this mask and the resulting binary image is treated as a matrix. The first row with a sum above a preset threshold is treated as the Y position of

the end of the magnet group; a similar procedure finds the X position. Surfaces of μ -magnets are inconsistent in color, so sometimes they will be almost entirely masked out. If this occurs with the end magnet, the X result can be labeled as unreasonable and replaced with an X value based on the previous detection. The isolation process also straightens out the group of magnets, so after the initial isolation the Y position is highly consistent. This information is sufficient to consistently tweeze the end μ -magnet.

C. Tweezer/Micromanipulator Calibration

An image processing scheme for detecting the tweezers is also employed. Using color thresholding and contour detection and extracting extreme points from the contours, the position of the tweezers and the jaw opening can be found, as shown in Fig. 8(A). The upper arm is relatively stationary during actuation, so its tip is found and declared as the tweezer position. However, if the tweezers are grasping something, it interferes with this detection. This is circumvented by detecting the object being grasped rather than the tweezers themselves – an example of this is shown in Fig. 8(B). Despite an imperfect estimation of the chassis, the algorithm is robust enough to accurately find its center. This calibration is advantageous because the chassis center will be moved to a desired location when (5) and (6) are applied instead of the tweezers tip position, which makes μ -magnet embedding more accurate.

V. ASSEMBLY RESULTS

Fig. 9 shows a set of 3 gripping robots assembled by AM μ RAS. They have been removed from the depot and washed. A single gripping robot can be fully assembled in approximately 311 seconds, which is 48% faster than the 600 seconds typically required for a well-practiced human. With the configuration presented in this paper, at most 10 grippers robots can be assembled before restocking μ -magnets on the μ -magnet stage would be required. This quantity can be raised to 30 by using a stage with three anchors, allowing three times as many μ -magnets to be stored at a time. Gripper assembly results are consistent and avoid damage to the chassis, which is a frequent issue in manual assembly. The correct orientation of μ -magnet poles is guaranteed by the automated assembly process.

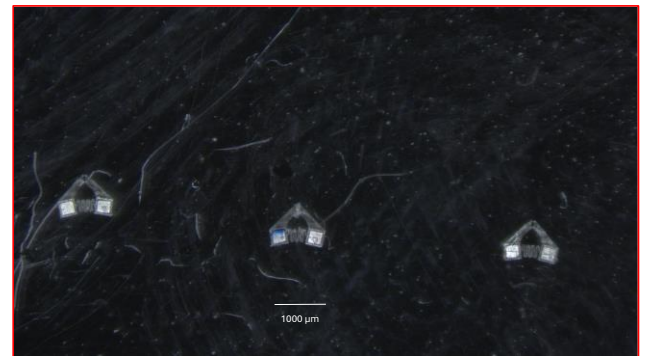


Figure 9. Set of three grippers assembled by AM μ RAS.

Fig. 10 shows two assembled tumbling robots on the robot depot, prior to washing or curing. The μ -magnets embedded are less visible due to being further from the top surface of the chassis. Each robot took about 192 seconds to assemble, which is 54% faster than the 420 seconds typically required for a well-practiced human. This, as well as the previous result for grippers, show that AM μ RAS is about twice as fast as a practiced human. The maximum number that can be assembled at once is again limited by the μ -magnet storage capacity. At most 20 robots could be made with the current configuration, and 60 by adding two more anchors. Consistent polishing and damage-free assembly are again guaranteed by the process. One issue is that μ -magnets are meant to be embedded deeper inside these chassis than those for grippers, i.e. the bottom surface of the magnet should be higher than the bottom of the chassis. A means by which to insert magnets deeper into a chassis would be a valuable future addition to this system. This could be accomplished by having a pin or similar object onto which the embedded magnet can be pushed, inserting it fully into the chassis.

VI. CONCLUSION

AM μ RAS is the first system to fully automate the assembly of micro-robots with embedded permanent magnets. This furthers progress towards scaling production quantity and makes feasible more micro-robot designs that can make use of the benefits offered by embedded permanent magnets. Further work to be done includes adding deep embedding capabilities, making a cleaner and more clinically viable system, and including more processes, such as liquid dispensing, in the automated assembly system.

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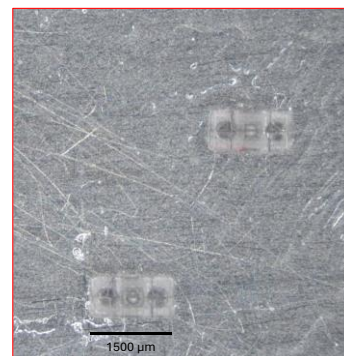


Figure 10. Two assembled tumbling robots on the chassis stage, prior to washing or curing. Embedded μ -magnets are less visible due to being further from the top surface of the robots.

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